

ChurnLens: Customer Risk Intelligence Platform

Comprehensive Project Documentation

1. Executive Summary

ChurnLens is an end-to-end Machine Learning pipeline and interactive React dashboard designed to predict customer churn and evaluate review integrity for e-commerce platforms. By combining transactional data (RFM metrics) with Natural Language Processing (NLP) on customer reviews, ChurnLens provides a holistic 360-degree risk profile for every customer.

2. The Real Problem We Are Solving

In the highly competitive e-commerce sector, acquiring a new customer is 5 to 25 times more expensive than retaining an existing one. The core problems businesses face are:

- Unpredictable Customer Attrition (Churn):** Companies often only realize a customer has left *after* they stop buying. There is no proactive intervention.
- Review Fraud and Mismatches:** Customers sometimes leave 5-star ratings but write scathing negative reviews (or vice-versa), polluting product metrics and hiding real sentiment.
- Siloed Data:** Transactional data (purchases) and behavioral data (reviews) are usually kept separate, preventing a unified view of customer risk.

The Solution: ChurnLens solves this by identifying *at-risk* customers before they leave using XGBoost, while simultaneously using NLP to detect sentiment anomalies in reviews, aggregating this into a single, actionable "Risk Score".

3. Technology Stack

This project is built using a modern, scalable, serverless-frontend architecture.

Data Engineering & Machine Learning (Python)

- Pandas & NumPy:** For heavy data manipulation, merging 100k+ rows across 9 tables.
- Scikit-Learn:** For data preprocessing, TF-IDF vectorization, and Logistic Regression.
- Imbalanced-Learn (SMOTE):** To handle the massive class imbalance between retained and churned customers.
- XGBoost:** The primary Gradient Boosting classifier used for predicting churn probability.
- VADER Sentiment:** For lexicon and rule-based sentiment analysis on customer reviews.

Frontend Application (JavaScript / React)

- React 18 & Vite:** Lightning-fast frontend framework and bundler.
 - Tailwind CSS v4:** Utility-first styling for a clinical, professional B2B SaaS aesthetic (Vercel/Datadog inspired).
 - Framer Motion:** For buttery-smooth, physics-based page transitions and animations.
 - Recharts:** For rendering dynamic, animated SVG charts.
 - PapaParse:** For in-browser CSV parsing, eliminating the need for a backend server.
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4. Step-by-Step Implementation Guide

How this project was built from scratch:

Phase 1: Data Engineering & Cleaning

- Loaded 9 relational tables from the Olist E-commerce dataset.
- Merged tables (Orders, Customers, Payments, Reviews, Products) using `customer_id` and `order_id`.
- Handled missing values (imputed medians for prices, filled missing text with placeholders).
- Converted timestamp strings to proper `datetime` objects.

Phase 2: Feature Engineering (RFM)

- Recency:** Calculated the days since the customer's last purchase.
- Frequency:** Counted the total number of unique orders per customer.
- Monetary:** Summed the total payment value per customer.
- Extra Features:** Extracted average delivery delays, average installments, and average review scores.

Phase 3: Churn Label Creation

- Defined a "Churned" customer as someone who has only ordered once AND hasn't ordered in the last 180 days.
- Created a binary target variable `churn` (1 = Churned, 0 = Retained).

Phase 4: XGBoost Predictive Modeling

- Split the engineered dataset into 80% Training and 20% Testing.
- Applied **SMOTE** (Synthetic Minority Over-sampling Technique) to the training data so the model doesn't just predict "Retained" every time.
- Trained an **XGBClassifier** to predict the churn probability.
- Evaluated using Precision, Recall, F1-Score, and AUC-ROC, exporting the `feature_importance.png`.

Phase 5: NLP Review Integrity Detection

- Vectorized customer review text using `TfidfVectorizer`.
- Calculated expected sentiment vs actual star rating.
- Trained a Logistic Regression model to classify whether a review's text matches its rating.
- Exported the `nlp_confusion_matrix.png`.

Phase 6: Frontend Dashboard Construction

1. Built a React SPA using Vite.
 2. Designed a pure dark-mode, high-contrast UI using Tailwind CSS.
 3. Created an interactive `dataLoader.js` to parse the processed CSVs directly in the browser.
 4. Built 5 distinct pages: Overview, Chum Analysis, Review Integrity, Customer Lookup, and Model Performance.
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5. SWOT Analysis

Strengths

- **Proactive rather than Reactive:** Identifies risk before the customer actually leaves.
- **Multi-modal Analysis:** Combines tabular numerical data with unstructured text data for better accuracy.
- **Serverless Architecture:** The React frontend parses CSVs directly, meaning zero hosting costs for a backend database.

Weaknesses

- **Static Data Dependency:** Currently, the dashboard reads from static CSV files. In a real-world enterprise scenario, it would need to be connected to a live SQL database via an API.
- **Language Limitations:** The current NLP model is primarily optimized for English text; localized e-commerce platforms would require multilingual embeddings.

Opportunities

- **Automated Marketing Hooks:** The system can be extended to automatically trigger discount emails via a CRM (like HubSpot or Salesforce) when a customer's Risk Score crosses 80%.
- **Deep Learning Upgrade:** Transitioning from XGBoost to neural networks (e.g., TabNet) if the dataset grows into the tens of millions of rows.

Threats

- **Data Privacy (GDPR/CCPA):** Handling granular customer data requires strict compliance. The system must ensure PII (Personally Identifiable Information) is anonymized.
 - **Concept Drift:** E-commerce behavior changes rapidly. A model trained on 2018 data will eventually degrade in accuracy in 2026 without continuous retraining.
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6. Conclusion

ChumLens transforms raw, messy e-commerce logs into a refined, predictive intelligence tool. By giving business stakeholders a sleek, Datadog-style interface to search for specific users and visualize global chum trends, ChumLens bridges the gap between complex data science and actionable business strategy.